WEATHER MODEL STUDY · DENMARK · 29 OCTOBER 2026

Blending eight weather models beats each of them, and beats the grid operator's forecast

A blend of eight public weather models forecast Danish wind power one day ahead better than any single model in 2026, and better than Energinet's own forecast. Which single model is best depends on whether you score it against a reanalysis or against real anemometers. The worst forecast day of the year was a market event, not a weather failure: the gusts were forecast right, the electricity was not.

DK2 POWER ERROR	BEST AT STATIONS	WIND CRPS	EASTER SUNDAY 2026
6.6%	**ICON-EU**	**0.50**	**30%**
blend, against 7.6% for Energinet	AIFS is first on ERA5, fifth at stations	m/s, neural blend; best single model 0.81	DK2 output in a gale the forecast got right; prices were negative

The question and the data

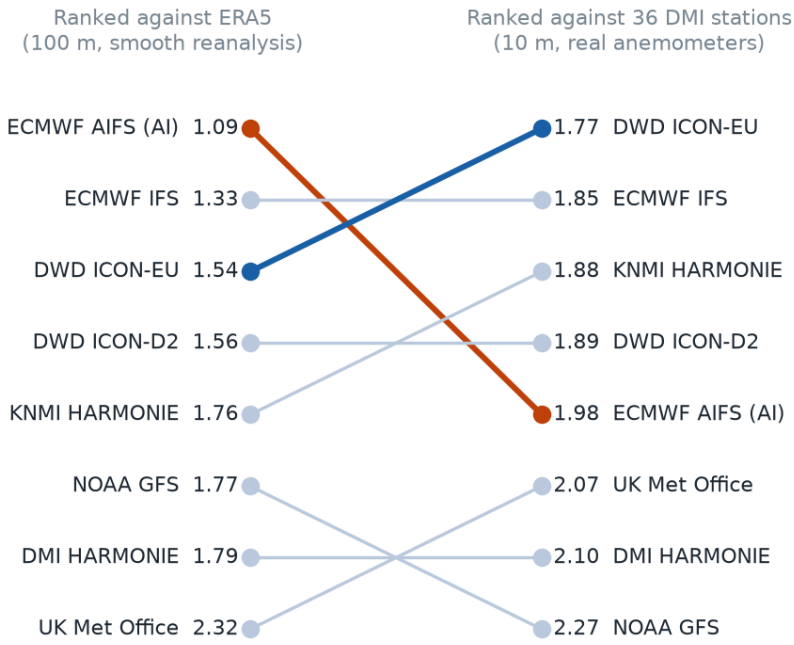
A power trader decides tomorrow's wind position before the 12:00 day-ahead auction, using weather forecasts issued that morning or the night before. This study asks which public weather model a Danish desk should trust, and what combining them buys.

Forecasts are Open-Meteo's archive of each model's run issued one day before the hour it forecasts, from March 2025 to October 2026. The eight models are ECMWF IFS and its AI model AIFS, DWD ICON-EU and ICON-D2, NOAA GFS, the UK Met Office, DMI HARMONIE and KNMI HARMONIE. They were sampled at 34 sites: the offshore farms and the main onshore clusters. Truth comes from ERA5, from DMI's open observations at 36 stations, and from Energinet's hourly wind production. Everything was trained on 2025 and tested on January to October 2026.

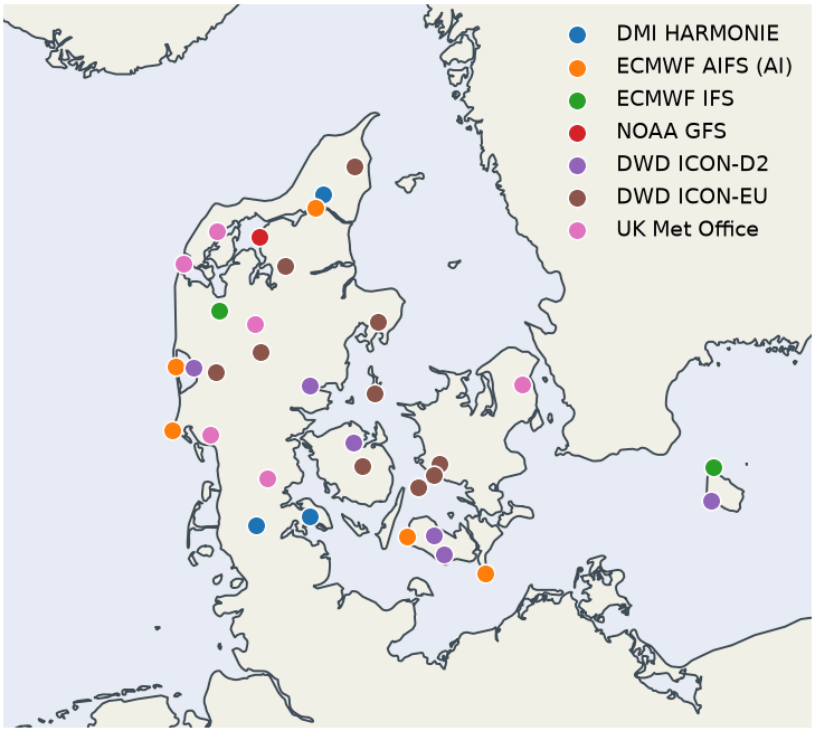
The best model depends on what you call the truth

Against ERA5 at 100 m, ECMWF's AI model AIFS has the smallest error of all eight. Against real anemometers at 10 m, it falls to fifth place and DWD's ICON-EU comes first.

ERA5 is a smooth reconstruction of past weather, and AIFS was trained to reproduce it, so AIFS is rewarded for smoothness. Anemometers see gusts and local effects that a smooth forecast misses. A desk that ranks models only against a reanalysis will pick a model that looks better than it is.



Day-ahead wind error (root mean square, m/s), March 2025 to October 2026. Left: against ERA5 at 100 m. Right: against 36 DMI stations at 10 m, each paired with a forecast site within 30 km.



The model with the smallest error at each DMI station. No model wins everywhere, which is the case for blending.

Blending beats every model and Energinet

Each model's 100 m winds at the 34 sites were turned into a power forecast for each price area by a gradient-boosted mapping. The blend is the plain average of the eight power forecasts. Error is given as a share of installed capacity.

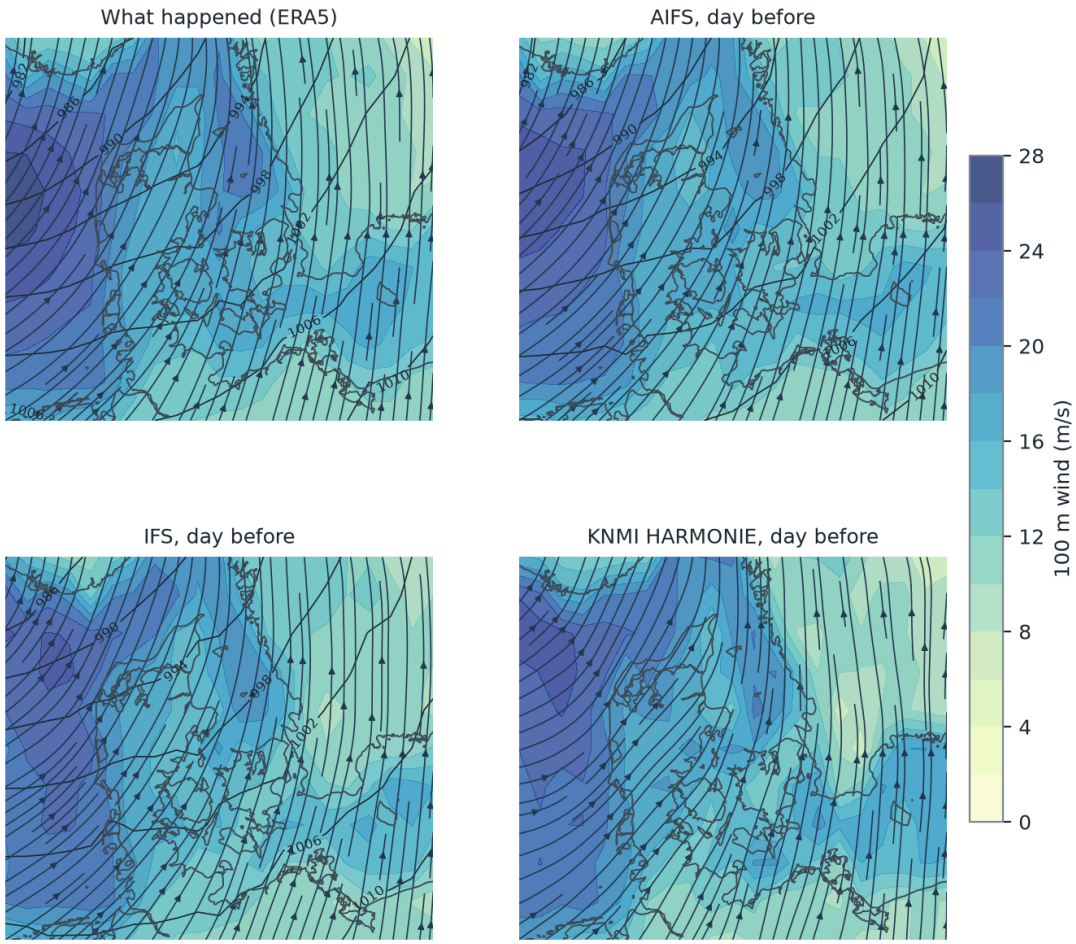
FORECAST, TEST JANUARY TO OCTOBER 2026	DK1	DK2
Blend, average of 8 models	5.7%	7.2%
Blend, one joint model	5.9%	6.6%
ECMWF AIFS alone (best single model)	6.3%	7.7%

FORECAST, TEST JANUARY TO OCTOBER 2026	DK1	DK2
ECMWF IFS alone	6.4%	8.3%
Energinet day-ahead, rescaled on 2025	6.5%	7.6%
Energinet day-ahead, as published	10.7%	8.4%

Mean absolute error as a share of capacity. The other six models fall between 6.7% and 7.4% in DK1. Energinet's DK1 offshore series is capped near 1,000 MW while the fleet peaks near 2,000 MW, so DK2 is the cleaner comparison.

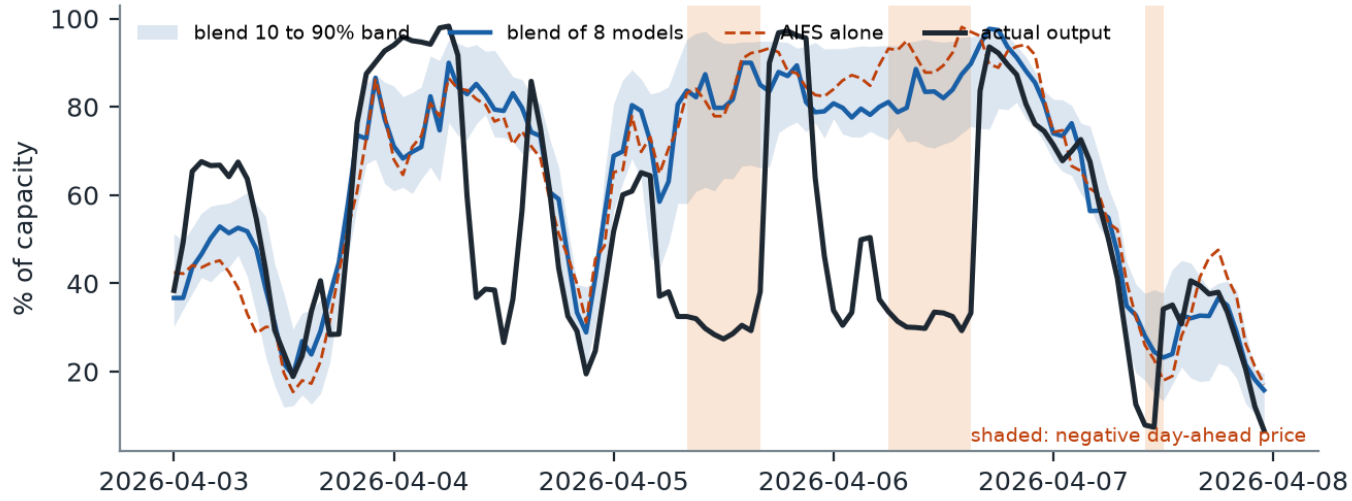
Easter 2026: the forecast was right and the turbines were switched off

The blend's worst days were 5 and 6 April 2026, Easter Sunday and Monday. A low crossed the North Sea, and the day-before forecasts placed it well.



100 m wind (colour), sea level pressure (lines, every 4 hPa) and flow (streamlines) at 06 UTC on 5 April 2026. ERA5 against the forecasts issued one day earlier. 0.5 degree grid.

Output did not follow the wind. Demand was low on the holiday, the day-ahead price turned negative, and wind farms were curtailed. DK2 output fell to about 30% of capacity in the middle of a gale that the forecast had right.



DK2 wind output, 3 to 7 April 2026, against the blend, its 10 to 90% band and AIFS alone. Shaded hours had a negative day-ahead price.

The gusts were right, the electricity was not

Above 15 m/s the blend's error rose to 14% of capacity in DK1 and 23% in DK2. The first suspect was turbines cutting out in high wind. Adding gusts, wind shear and a cut-out flag changed the gale error by less than one point, so that suspect was cleared.

The cause was the market. In DK2 gale hours with a negative price, the blend over-forecast by about half of capacity, while in gale hours with a positive price it was nearly unbiased. The price would fix most of it, but it is set at the 12:00 auction, after the 10:30 forecast, so using it in the day-ahead forecast would be look-ahead. After the auction it is public, and for intraday trading feeding it to the model cut the DK2 error in those hours from 52% to 14% of capacity.

Before the auction, the desk has to predict negative prices itself. A first classifier using only 10:30 information (the eight models' wind, solar and wind forecasts for Denmark and Germany, holidays, and today's prices) ranked the risky hours well, with an AUC of 0.93 in DK1. But negative hours were rarer in 2026 than in 2025, its probabilities stayed too low to act on, and it did not reduce the gale error. Recalibrating it on recent months is the next step.

DK2, TEST 2026	ALL HOURS	GALE, ANY PRICE	GALE, PRICE AT OR BELOW ZERO
Blend	7.1%	22.8%	51.6%
Plus gust, shear and cut-out	7.0%	22.0%	42.4%
Plus the day-ahead price (intraday only: known after the auction)	6.2%	9.5%	13.5%

Gale: hours when ERA5 100 m wind averaged over the DK2 sites exceeds 15 m/s. Only 21 gale hours had a price at or below zero, so the last column rests on few hours.

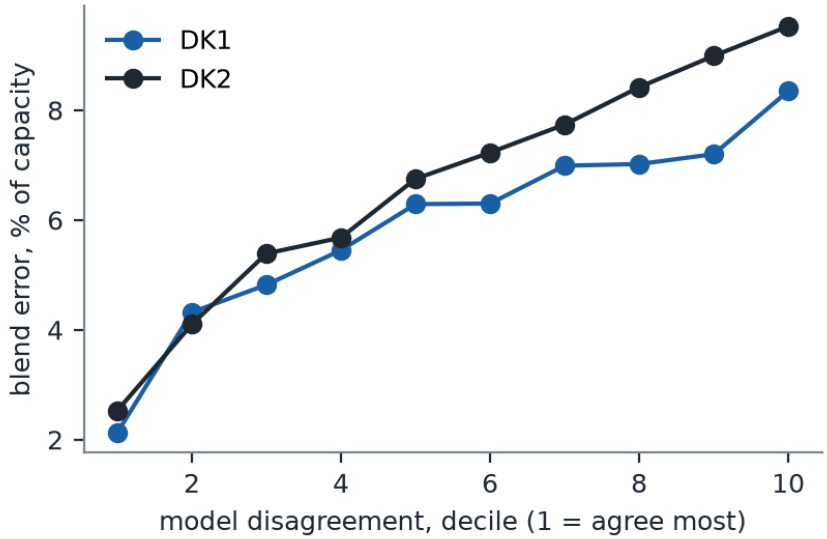
An uncertainty band that holds its promise

Three post-processing methods from the forecasting literature turn the eight models into a probability distribution for 100 m wind. EMOS fits a calibrated normal distribution to the models' mean and spread. DRN (Rasp and Lerch, 2018) and Bernstein quantile networks (Bremnes, 2020) learn the distribution from all 57 inputs with a small neural network. All three beat the best single model by a wide margin on the continuous ranked probability score (CRPS), which rewards a forecast that is both sharp and honest.

100 M WIND, TEST 2026	CRPS, M/S
Bernstein quantile network	0.50
DRN neural network	0.51
EMOS, fitted per site	0.54

100 M WIND, TEST 2026	CRPS, M/S
The eight models as a raw ensemble	0.62
Best single model, AIFS	0.81

For power, the first quantile model was too confident: its 10 to 90% band caught only half the outcomes. A conformal correction (Romano and others, 2019) widened it on held-out 2025 data, and the band then caught 86% of outcomes in DK1 and 83% in DK2, close to the 80% it promises.



When the eight models disagree most, the blend's error is about twice as large as when they agree. Disagreement works as a warning light. A separate test found it does not predict the direction of the error, so it is a risk signal, not a trading signal.

Running forecast

A live version issues tomorrow's forecast every day at 10:30, before the 12:00 auction, and scores it afterwards. The record is kept in git with timestamps. On its July to October 2026 backtest, AIFS and IFS tie for power once each has its own mapping. Pushing AIFS winds through a mapping built for IFS under-forecasts by about 40%, because AIFS winds are smoother.

Limits

The test covers ten months with one train and test split and only three strong storms, so rankings a few tenths of a point apart are ties. ERA5 flatters smooth models, which is why every model is also scored against stations. Installed capacity is estimated from each area's 60-day peak output, because no public hourly capacity series exists while Thor was coming online. The storm maps use a 0.5 degree grid, which smooths the high-resolution models.

Bolgaç Gülen, 29 October 2026. Data: Open-Meteo forecast archive, ERA5 via Open-Meteo, DMI open data, Energinet Energi Data Service, Natural Earth coastlines. Interactive version with animated wind maps: bolgacg.github.io/dk-wind-blend.